Data Management, Winter 2026/27 · Group project

From raw race results to a FAIR, quality-assured knowledge graph

Formula 1 World Championship results 1950–2025

Student A
FAIR assessment and metadata design
Chirag Adhikari

Student B
Data quality and profiling
Manohar Zalki

Student C
Data cleaning and pipeline
Nicolas Bonilla

76 seasons of F1 results with real quality issues

25,907
race results (one row per car per race)

1,149
championship Grands Prix

817
drivers and 203 constructor entries

77
circuits in 34 countries

Why this dataset

- Relational: one fact table and four reference tables
- Genuine issues from 76 seasons of rule changes and file exports
- A natural network of drivers, teams, circuits and countries

Source: Ergast database via the Jolpica-F1 API (also mirrored on Kaggle). Sprint races are out of scope.

Presenter: Student A · Chirag Adhikari

One command rebuilds every deliverable

Step 0
Extract
25,907
raw rows, values unchanged

→

B
Profile
29
issues in 7 quality dimensions

→

C
Clean
55/55
validation gates passed

→

A
Describe
54→93%
RDA FAIR maturity

→

Graph
Model
30,141
typed relationships

→

Web
Show
7
dashboard views, live

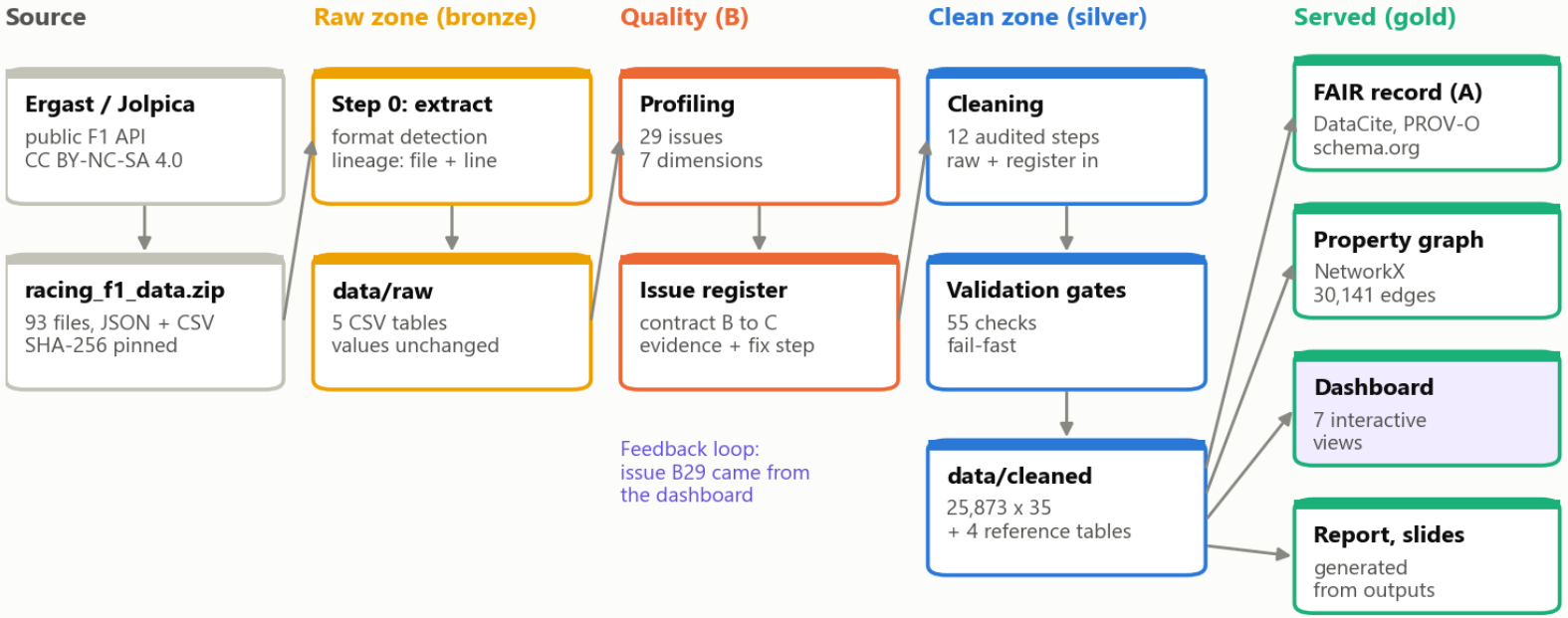
```
python run_all.py --from-scratch
```

Deterministic: two runs give byte-identical files · 14 automated tests · no network needed

Data engineering: zones, contracts and controls

End-to-end data flow: from the raw archive to the dashboard

Every arrow is code in run_all.py; each zone reads only the zone before it.



Cross-cutting engineering controls

Schema as code
schema.py, vocabularies.py:
one contract for C and A

Orchestration
run_all.py: one command,
byte-identical reruns

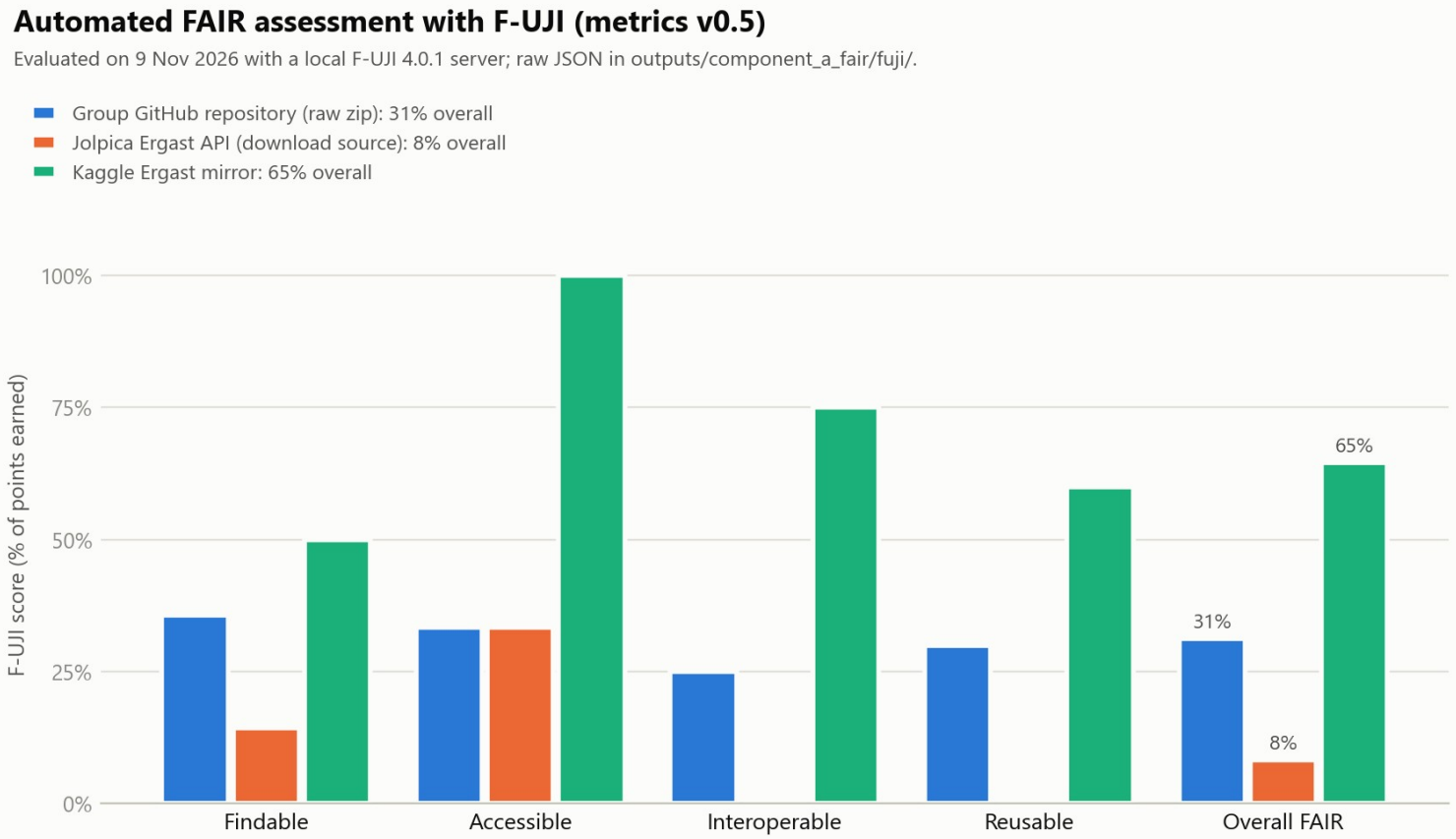
Automated tests
pytest on key decisions
(order, keys, sentinels)

Lineage
file + line per row,
checksums, PROV-O

Numbers read from the generated outputs; group project, Data Management 2026/27.

Presenter: Student A · Chirag Adhikari

As obtained, the data is hard to find and reuse



54% → 93%

RDA maturity, 41 indicators: as obtained → our package

6/20 → 12/20

essential indicators met

Findable stays at level 0 until a DOI is actually minted.

Presenter: Student A · Chirag Adhikari

Biggest gaps: no identifier, conflicting licences

Who says what about the licence?

Ergast (original source)

Non-commercial use only

Jolpica-F1 (our download)

CC BY-NC-SA 4.0

Kaggle mirror

Claims CC0, public domain

Our GitHub repository

MIT licence for everything

Fix: data under CC BY-NC-SA 4.0, code under MIT, with attribution

What our package adds

98.7% — of drivers linked to Wikidata IDs (circuits 100%)

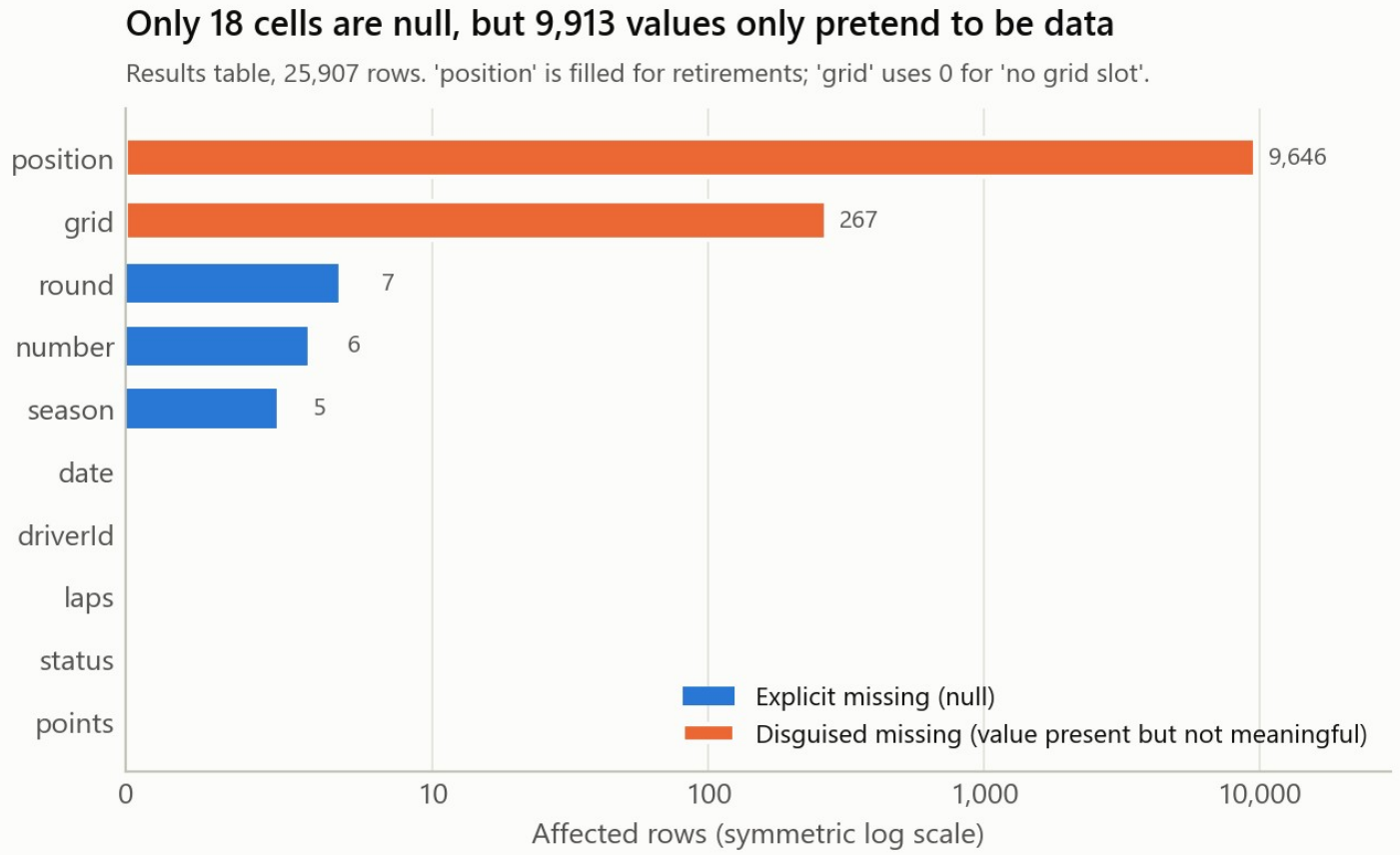
DataCite 4.5 — record validated against the official XSD

ISO + PROV — ISO 3166 country codes, W3C provenance, checksums

DOI — via Zenodo: reserved now, minted on deposit

Presenter: Student A · Chirag Adhikari

Completeness is deceptive: 9,913 hidden gaps



18

explicit nulls: the table looks 99.9% complete

9,646

'positions' that are really retirements

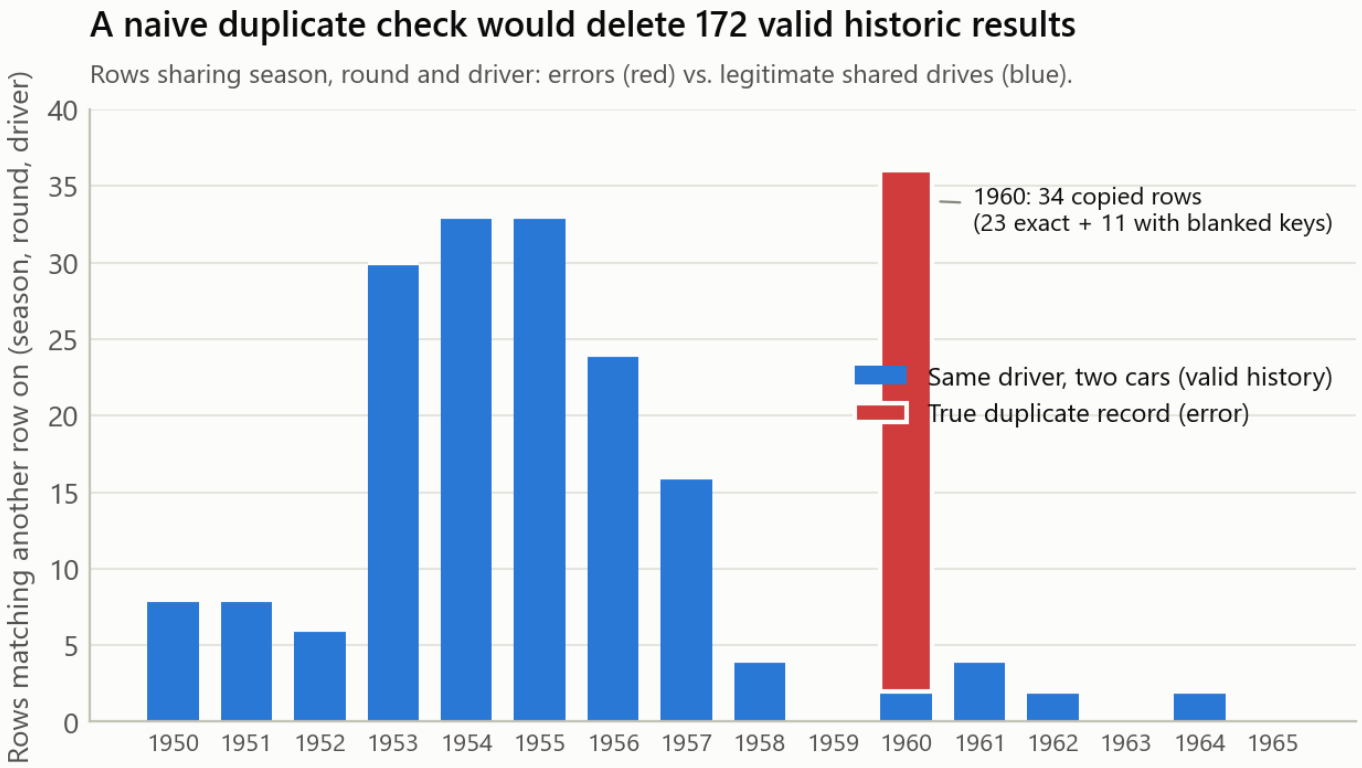
267

cars on a fictitious 'grid slot 0'

Source: Ergast/Jolpica F1 results 1950-2025 (racing_f1_data.zip); group analysis.

Presenter: Student B · Manohar Zalki

A naive duplicate check deletes real history



Source: Ergast/Jolpica F1 results 1950-2025 (racing_f1_data.zip); group analysis.

34

true duplicates, all from 1960

11

of them hidden behind blanked keys

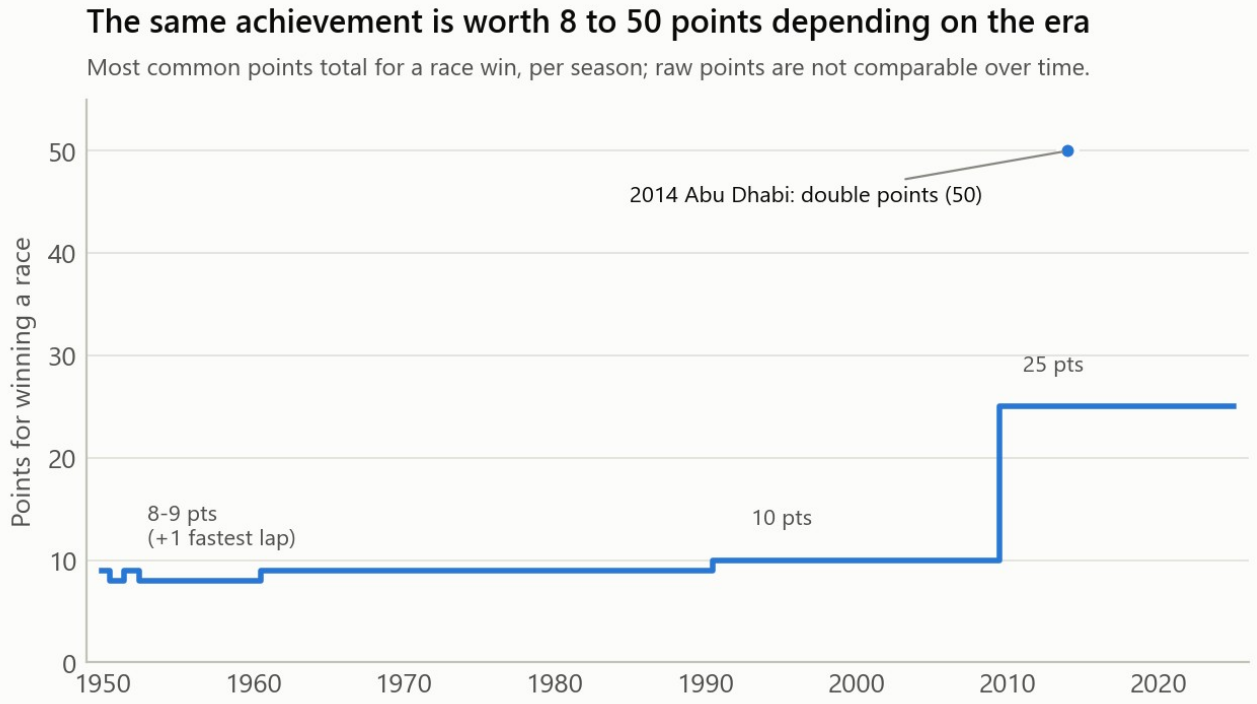
172

valid rows: one driver, two cars in a race

Correct key: season, round, driver and car number

Presenter: Student B · Manohar Zalki

Valid history can look like bad data



Source: Ergast/Jolpica F1 results 1950-2025 (racing_f1_data.zip); group analysis.

8–50

points for the same achievement: winning a race

270

lap 'outliers', all valid Indianapolis 500 results: flag, never delete

Impact on data science and AI

- Inflated wins from duplicated rows
- Biased targets: a retirement as '18th'
- Era bias in all-time rankings

Presenter: Student B · Manohar Zalki

Order matters: repair keys, then de-duplicate

Naive order

1. Remove exact duplicates
2. Repair missing keys

11

duplicates survive, hidden by the blanked keys

Our order (steps C02 → C03)

1. Restore season (5) and round (7) by exact lookup
2. Remove exact duplicates

34

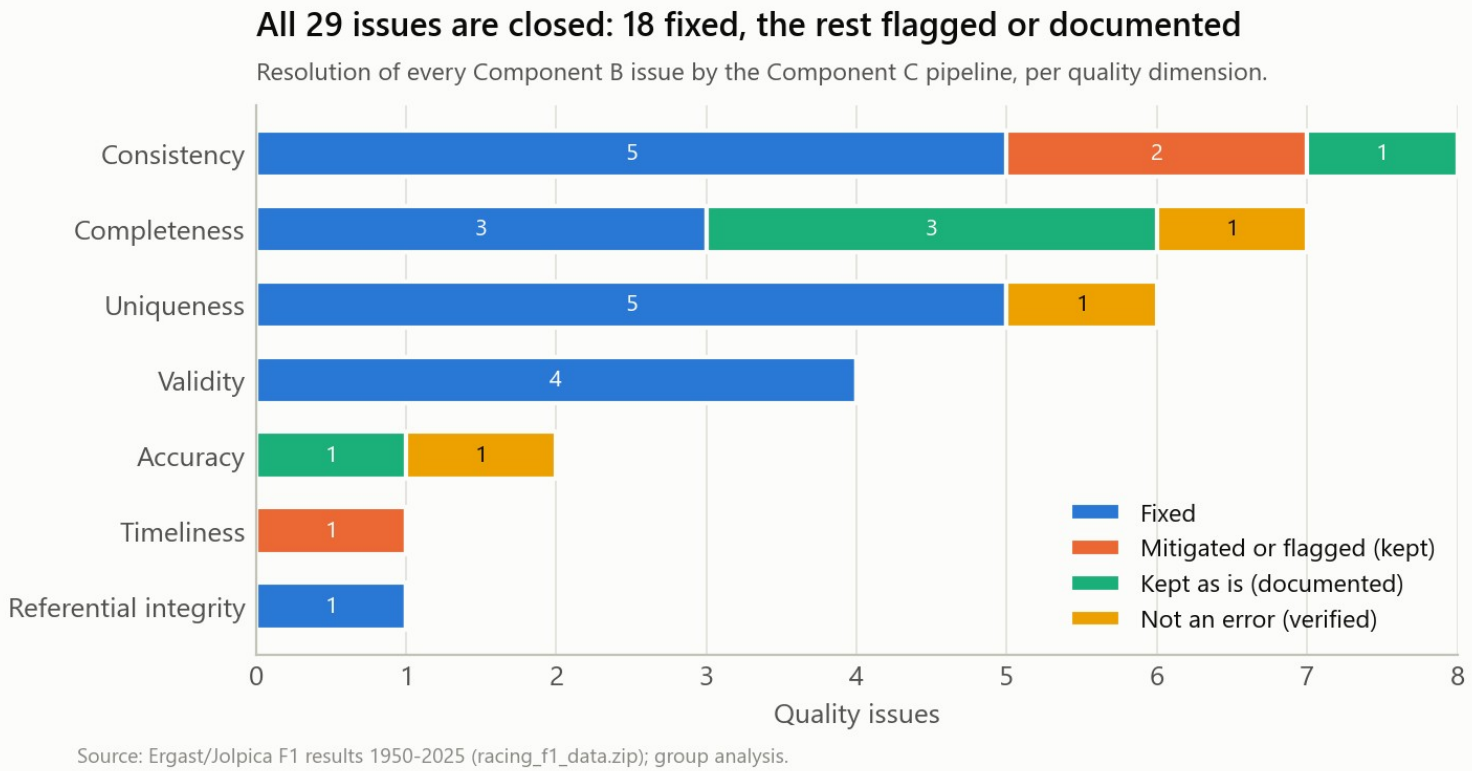
duplicates removed, none left; a unit test locks this in

No invented values

- Grid 0 becomes missing plus a flag (267 rows); no statistical imputation anywhere
- Keys come only from deterministic lookups: the file name and races.csv

Presenter: Student C · Nicolas Bonilla

Every issue closed, every output validated



Source: Ergast/Jolpica F1 results 1950-2025 (racing_f1_data.zip); group analysis.

- Flag, don't delete: Indy 500 rows, shared drives and status conflicts stay, with flags

55/55

validation gates passed; one failure blocks publishing

18 of 29

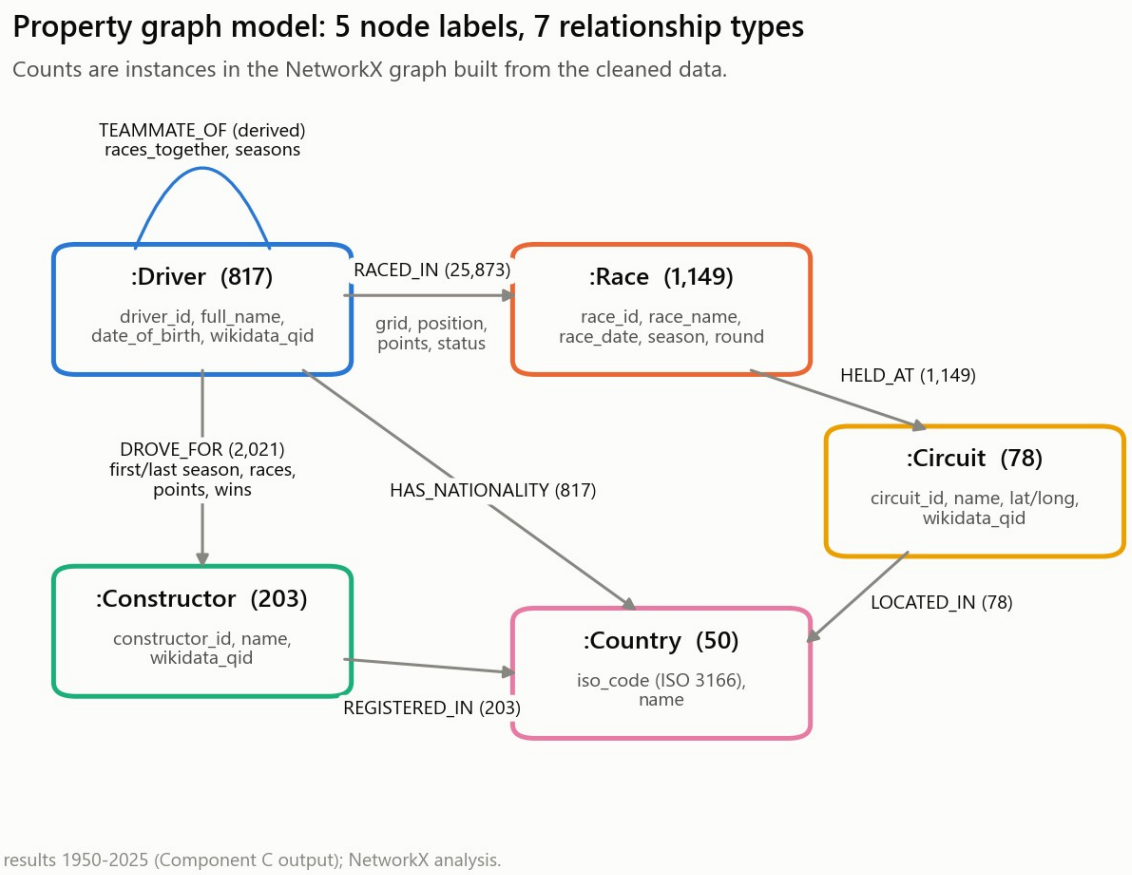
issues fixed, the rest flagged, kept or verified

25,873

clean rows, 35 documented columns, an audit record per step

Presenter: Student C · Nicolas Bonilla

75 years of F1 form a small world



Source: cleaned F1 results 1950-2025 (Component C output); NetworkX analysis.

NetworkX MultiDiGraph: 2,297 nodes, 30,141 typed edges, exported to GEXF for Gephi

0.41

clustering (random graph: 0.014)

4.8

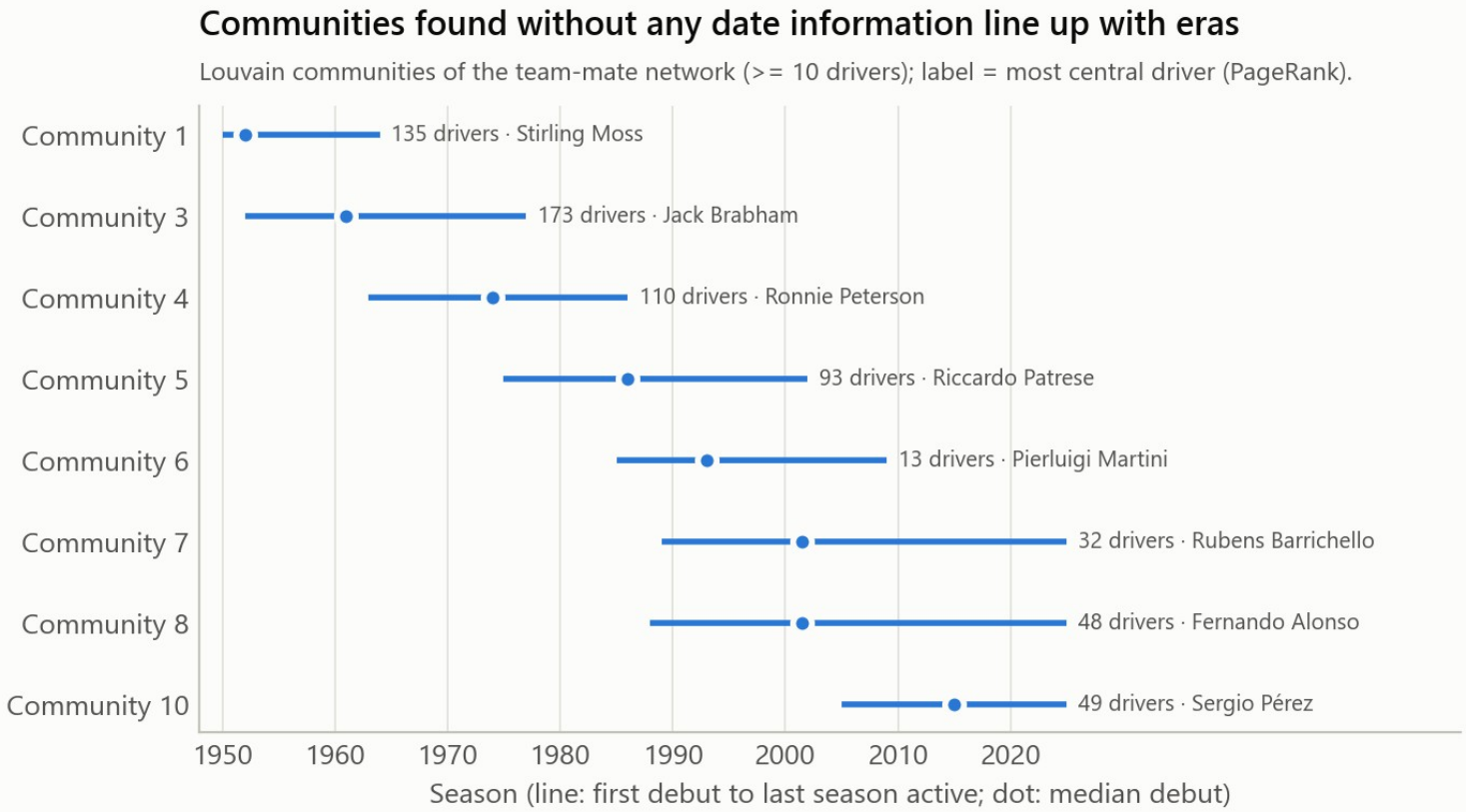
team-mate links between two drivers on average

Fangio → Verstappen in 8 steps

Fangio → Bonnier → Andretti → Giacomelli → de Cesaris → Barrichello → Button → Pérez → Verstappen

Presenter: Student B · Manohar Zalki

Communities rediscover the eras by themselves



Source: cleaned F1 results 1950-2025 (Component C output); NetworkX analysis.

11

Louvain communities, modularity 0.70

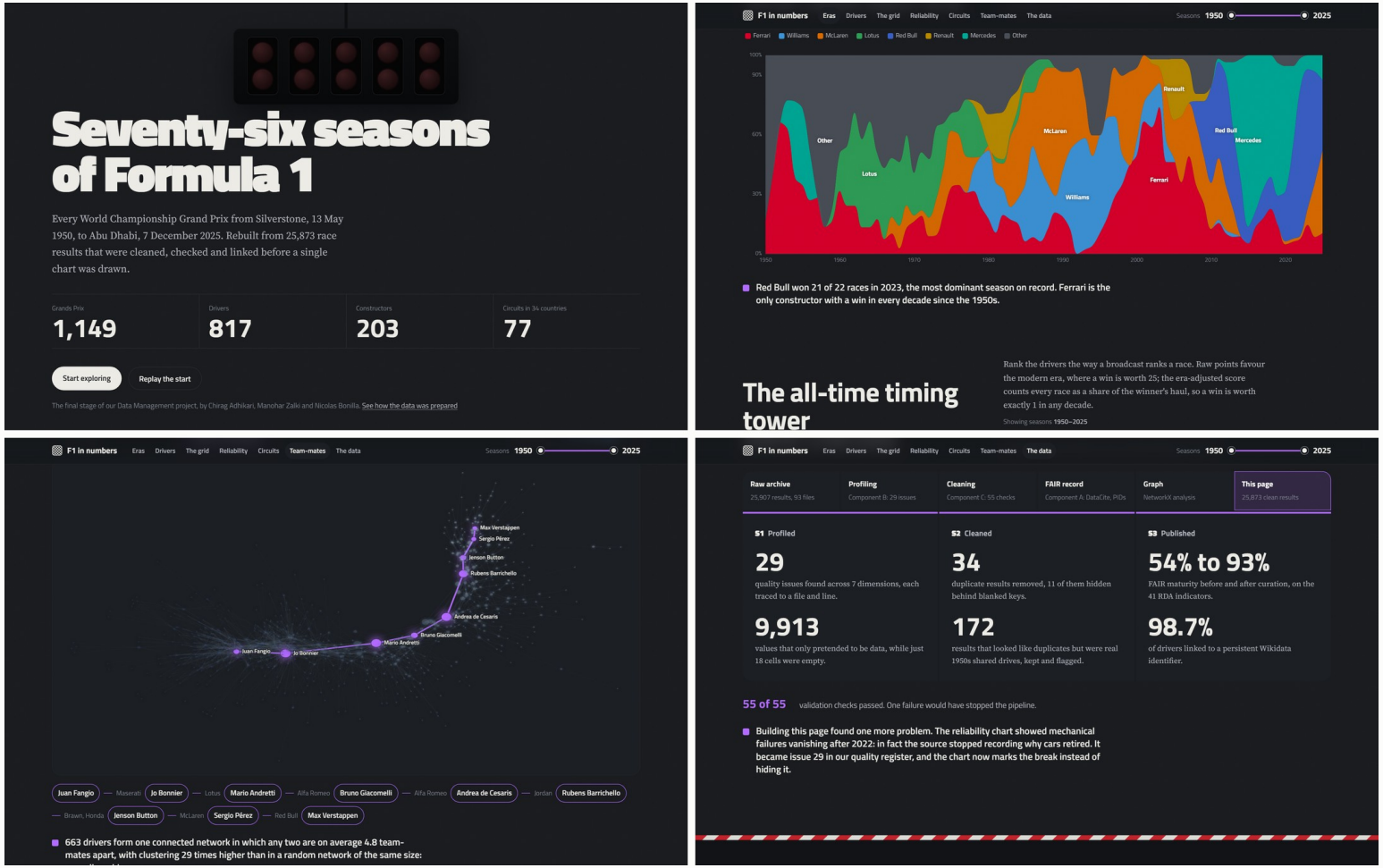
0.15

highest betweenness; top bridges: Bonnier, Brabham, de Cesaris

Caveat

Before the 1970s 'constructor' means the chassis make that private entrants shared, which inflates early drivers' degree.

The project, live: our interactive dashboard



Built only from validated outputs

- build_web_data.py reads the clean tables, graph metrics, issue register and FAIR scores
- run_all.py rebuilds it as its last step
- It found issue B29: retirement causes not recorded from 2023

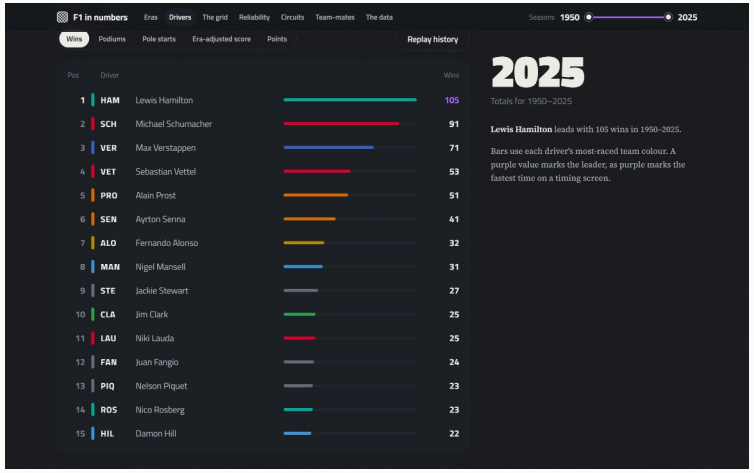
Live demo

```
streamlit run app.py
```

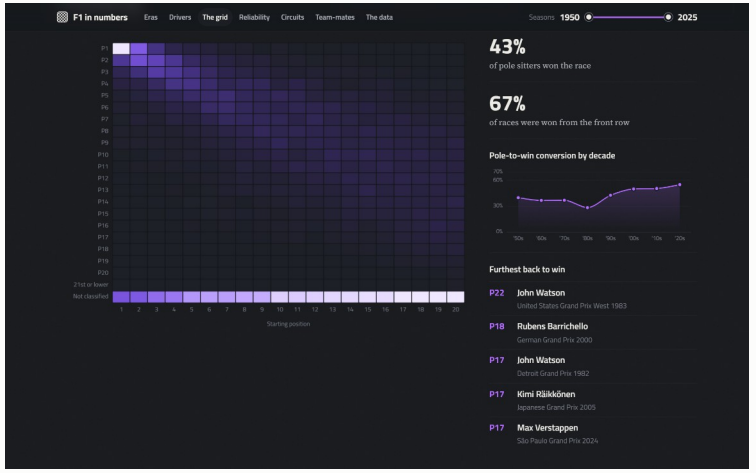
then localhost:8501

Presenter: Student C · Nicolas Bonilla

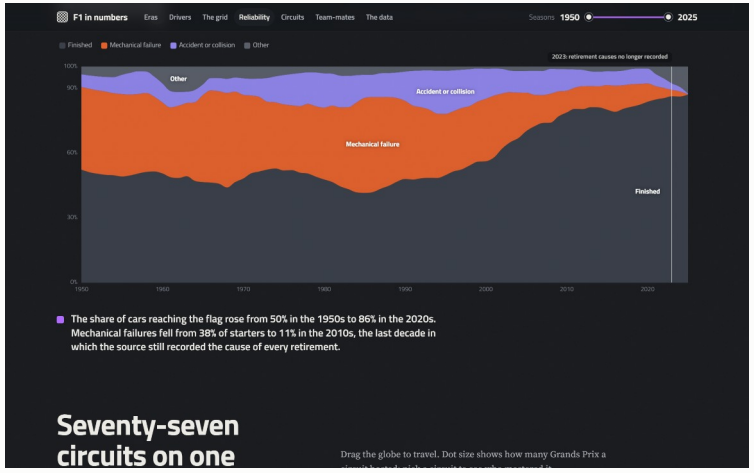
Inside the dashboard: every chart reads validated data



Timing tower: wins, podiums, era-adjusted score



Grid to finish, pole conversion, furthest-back wins



Reliability, with the 2023 break (B29) marked



77 circuits on a rotating globe

What we learned

1

Quality is contextual

Outliers, duplicates and short grids were often valid history

2

Order of operations matters

11 duplicates hid behind blanked keys

3

FAIR gaps sit around the values

No DOI, conflicting licences, no provenance

4

Automate the evidence

Audit log, 55 gates, 14 tests, metadata generated from code

Questions?

Code, data, metadata and report: github.com/zalkimanohar/Formula1-Racing-Analytics

Rebuild everything: `python run_all.py --from-scratch`

Student A · Chirag Adhikari

Student B · Manohar Zalki

Student C · Nicolas Bonilla